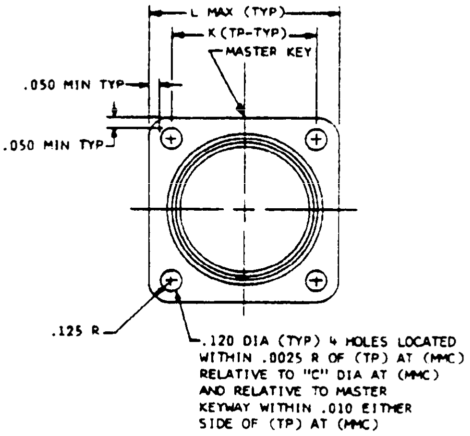
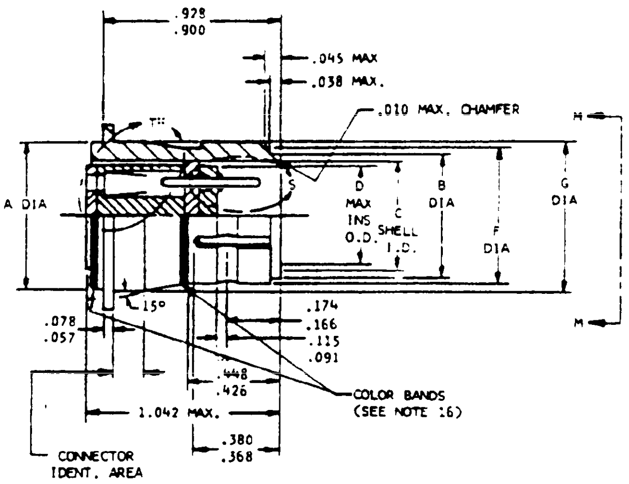
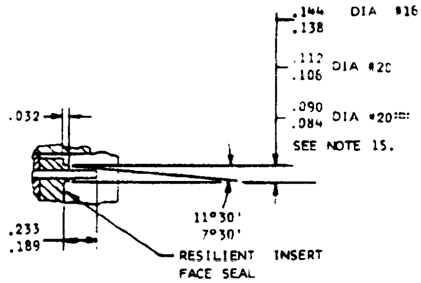
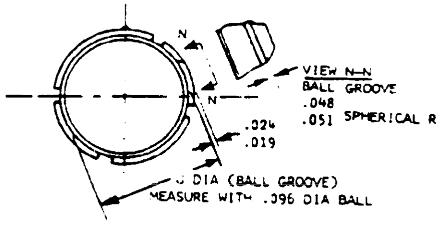
FED SUP CLASS
5935


L MAX (TYP)
K (TP-TYP)
MASTER KEY
.050 MIN TYP
.050 MIN TYP
.125 R
.120 DIA (TYP) 4 HOLES LOCATED WITHIN .0025 R OF (TP) AT (MMC) RELATIVE TO "C" DIA AT (MMC) AND RELATIVE TO MASTER KEYWAY WITHIN .010 EITHER SIDE OF (TP) AT (MMC)


.928
.900
.045 MAX
.038 MAX.
.010 MAX. CHAMFER
T::
S
A DIA
D MAX INS O.D.
C SHELL I.D.
B DIA
F DIA
G DIA
M
M
15°
.078
.057
.174
.166
.115
.091
.448
.426
1.042 MAX.
.380
.368
COLOR BANDS (SEE NOTE 16)
CONNECTOR IDENT. AREA

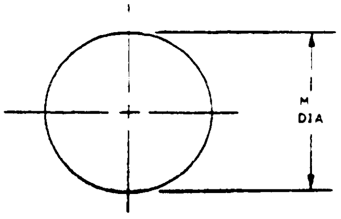

.144
.138
DIA #16
.112
.106
DIA #20
.090
.084
DIA #20:::
SEE NOTE 15.
.032
.233
.189
11°30'
7°30'
RESILIENT INSERT FACE SEAL

DETAIL S


N
N
VIEW N-N
BALL GROOVE
.048
.051 SPHERICAL R
.024
.019
J DIA (BALL GROOVE)
MEASURE WITH .096 DIA BALL

VIEW M-M

FIGURE 1
CONNECTOR VIEW



PANEL CUTOUT

:: SEE SHEET 2 FOR DETAIL T

::: ON OUTER ROW CONTACTS OF INSERT ARRANGEMENTS INDICATED IN CHART

PART NUMBER EXAMPLE

MS3145 N 3 - 50 P Y
- Y — ALTERNATE INSERT POSITION
- P — CONTACT STYLE (P-PIN)
- 50 — INSERT ARRANGEMENT
- 3 — SHELL SIZE
- N — CLASS
- MS3145 — MS NUMBER

INSERT ARRANGEMENTS REQUIRING REDUCED DIAMETERS FOR PIN CONTACT RAISED SEAL BARRIERS

SHELL SIZE	INSERT ARRANGE-MENTS	CONTACT CAVITIES
12	12-50	5, 6, 8, 9, 11 & 12

APPROVED 4 Feb 1972 REVISED

Review activities:
Navy-EC

This standard has been approved by the Department of the Navy and is mandatory for use by that activity. All other military activities are requested to employ this standard where suitable

P.A. Navy-AS
Other Cust

TITLE
CONNECTOR, RECEPTACLE, ELECTRIC, HERMETIC, CRIMP TYPE, BOX MOUNTING, PUSH-PULL COUPLING, SERIES 3 (CLASS N)

MILITARY STANDARD

MS3145(NAVY)

PROCUREMENT SPECIFICATION
MIL-C-81703

SUPERSEDES

SHEET 1 OF [illegible]

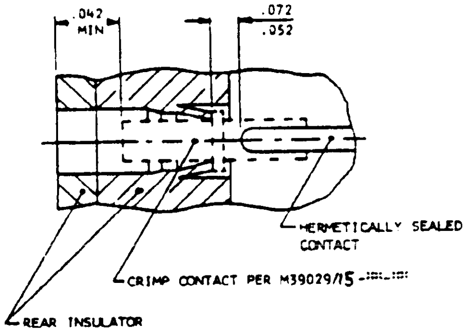
DD FORM 672-1 N

LIMITED COORDINATION

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

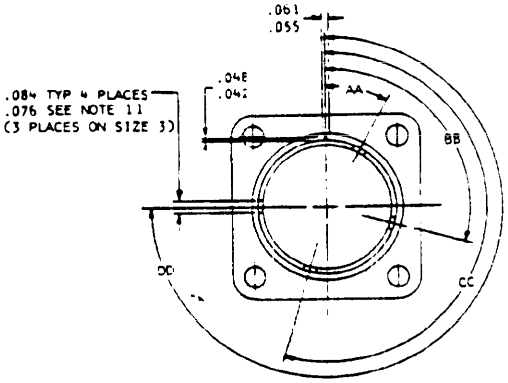
Project No. 5935-N668-3

FED SUP CLASS
5935



.042
MIN
.072
.052
HERMETICALLY SEALED CONTACT
CRIMP CONTACT PER M39029/15
REAR INSULATOR

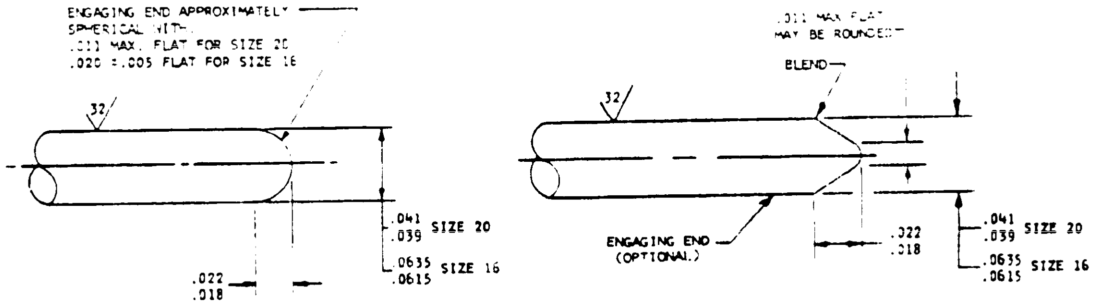
DETAIL T



.061
.055
.084 TYP 4 PLACES
.076 SEE NOTE 11
(3 PLACES ON SIZE 3)
.048
.042
AA
BB
CC
DD

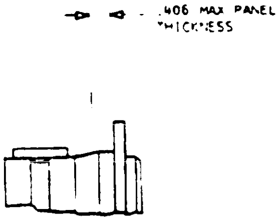
FIGURE II
KEYWAY POSITIONS

ENGAGING ENDS OF CONTACTS



ENGAGING END APPROXIMATELY SPHERICAL WITH .011 MAX. FLAT FOR SIZE 20 .020 ±.005 FLAT FOR SIZE 16
32
.041
.039
SIZE 20
.0635
.0615
SIZE 16
.022
.018
.011 MAX FLAT MAY BE ROUNDED
BLEND
32
ENGAGING END (OPTIONAL)
.022
.018
.041
.039
SIZE 20
.0635
.0615
SIZE 16

FIGURE III
CONTACT DETAILS



.406 MAX PANEL THICKNESS

FIGURE IV
MAX PANEL THICKNESS (TYP)

Review activities:
Navy-EC

APPROVED 4 Feb 1972 REVISED

This standard has been approved by the [illegible] of the Navy and is mandatory for use by [illegible]. All other military activities are required to [illegible] this standard where suitable.

P A Navy-AS Cther Cust	TITLE CONNECTOR, RECEPTACLE, ELECTRIC, HERMETIC, CRIMP TYPE, BOX MOUNTING, PUSH-PULL COUPLING, SERIES 3 (CLASS N)	MILITARY STANDARD MS3145(NAVY)
PROCUREMENT SPECIFICATION MIL-C-81703	SUPERSEDES	SHEET 2

DD FORM 1 APR 72 672-1 N LIMITED COORDINATION PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

FED SUP CLASS
5935

TABLE I
CONNECTOR DIMENSIONS AND WEIGHTS

SHELL SIZE	A	B	C	D	F	G	J	K	L
3	.497	.392 .385	.367 .356	.330	.441 .431	.531	.398 .391	.625	.391
7	.622	.527 .520	.483 .473	.451	.576 .566	.671	.533 .526	.719	1.016
12	.747	.661 .654	.624 .614	.553	.710 .700	.812	.668 .661	.813	1.109
19	.904	.800 .793	.763 .753	.738	.849 .839	.937	.806 .799	.906	1.203
27	1.060	.955 .948	.918 .908	.854	1.004 .994	1.125	.961 .954	.963	1.297
37	1.153	1.077 1.070	1.040 1.030	.976	1.126 1.116	1.218	1.083 1.076	1.187	1.453

SHELL SIZE	M	WEIGHT (LBS) SEE NOTE 7 PIN INSERT
3	.555	(WEIGHT WILL BE SUPPLIED IN LATER REVISION)
7	.688	
12	.828	
19	.953	
27	1.136	
37	1.234	

TABLE II
KEYWAY POSITIONS

SIZE	AA	BB	CC	DD
3	23°	97°	187°	--
7	23°	97°	187°	277°
12	30°	105°	195°	270°
19	30°	105°	195°	270°
27	30°	105°	195°	270°
37	30°	105°	195°	270°

Review activities:
Navy-EC

NOTES: (NOTES 1 THRU 6 APPLICABLE TO CONTACTS ONLY).

1. CONTACT FINISH: GOLD PLATE .000050 INCH MIN. IN ACCORDANCE WITH MIL-G-45204. A SUITABLE UNDERPLATING MAY BE USED. SILVER UNDERPLATING SHALL NOT BE USED. FINISH IS APPLIED AFTER SEALING.
2. DIMENSIONS IN INCHES, DIMENSIONS APPLY AFTER PLATING.
3. 63✓ ALL OVER PER MIL-STD-10, UNLESS OTHERWISE SPECIFIED.
4. REMOVE ALL BURRS AND SHARP EDGES .003 TO .005 R.
5. DIAMETERS TO BE CONCENTRIC WITHIN .004 TIR.
6. DIAMETERS IN THE SAME CROSS-SECTIONAL PLANE TO BE CONCENTRIC WITH EACH OTHER WITHIN .003 TIR.
7. WEIGHT LISTED IS CONNECTOR MAXIMUM WEIGHT PER SHELL SIZE LESS CRIMP CONTACTS.
8. SHELL MATERIAL: COLD ROLLED STEEL, PER QQ-S-637
9. SHELL FINISH: .0001 MIN. TIN PER MIL-T-10727, OVER NICKEL PER QQ-N-290.
10. EACH CONNECTOR SHALL BE PACKAGED WITH A PLASTIC DUST CAP OVER EACH END OF SHELL.
11. FOUR KEYWAYS (MMC) AND INSERT SHALL BE LOCATED WITHIN .004 EITHER SIDE OF (TP) RELATIVE TO MASTER KEYWAY (MMC) AND SHELL O.D. (MMC).
12. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE SHALL BE .XX ± .010 AND .XXX ± .005; ANGULAR TOLERANCE, X° ±1 -0' AND X°X' ± 30'.
13. INSERTION/REMOVAL TOOL, MS3160-[illegible] TO BE USED WITH THIS CONNECTOR.
14. CONTACT MIL-C-39029/15-[illegible] MUST BE USED FOR WIRE TERMINATIONS.
15. DIAMETERS SHOWN ARE AT BASE OF BARRIER.
16. COLOR BAND: (A) .070 ±.031 WIDE, COLOR BLUE.
(B) BAND LOCATION MUST BE SUCH THAT IT IS VISIBLE WHEN MOUNTED.
17. MAXIMUM WIRE DIAMETER PERMISSIBLE: .074 FOR SIZE 20. .103 FOR SIZE 16 CONTACT.

APPROVED 4 Feb 1972 REVISED

This standard has been approved by the Department of the Navy and is mandatory for use by the Navy activity. All other military activities are requested to employ this standard where suitable

P.A. Navy-AS Other Cust	TITLE CONNECTOR, RECEPTACLE, ELECTRIC, HERMETIC, CRIMP TYPE, BOX MOUNTING, PUSH-PULL COUPLING, SERIES 3 (CLASS H)	MILITARY STANDARD MS3145(NAVY)
PROCUREMENT SPECIFICATION MIL-C-81703	SUPERSEDES	SHEET 3

DD FORM 1 APR 67 672-1 N   LI [illegible] CORPORATION   PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE